

ZHEYE SONG

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EDUCATION

Fudan University, 87/100

B.S. in Mathematics and Applied Mathematics

Sept 2022 – Jun 2026

The Hong Kong Polytechnic University, 3.97/4.3

Exchange Student

Sept 2024 – Jan 2025

RESEARCH INTERESTS

Causal Inference, Policy Learning, and Statistical Learning for Complex Data.

RESEARCH PROJECTS

Causal Inference in Metric Spaces via Distance Profiles

Advisor: Zhiheng Zhang, Tenure-Track Assistant Professor, SUFE

2025.08 – 2026.05

Under Review

- Developed a distance-profile causal inference framework for object-valued outcomes by transforming metric-ball events in separable metric spaces into geometry-indexed binary potential-outcome contrasts.
- Proved pointwise identification, doubly robust/AIPW estimation properties, Neyman orthogonality, and auxiliary-stabilized finite-grid simultaneous inference under causal identification and high-level nuisance conditions.
- Validated the framework across Euclidean, Wasserstein, SPD, graph, and tree-valued outcomes, with semi-synthetic connectome and network benchmarks supporting calibrated coverage, global testing, and robustness diagnostics.

Modular Causal Meta-Learners for CATE Estimation

Advisor: Jiayi Yang, Alumni Researcher, Columbia University

2025.04 – 2026.04

Research Project

- Built a BART-based cross-fitted nuisance layer and modular CATE estimator library, separating S-, X-, R-, and BCF-inspired correction principles under a common implementation.
- Implemented HybridFull, an overlap-aware orthogonal-residual stacking rule that learns simplex weights over candidate CATE estimators within propensity strata, and evaluated it across S1–S6 simulations plus IHDP/JOBS benchmarks.

Relative-Belief Multiple Testing under General Additive Loss

Collaborator: Qiaoyu Liang, Ph.D. student, University of Toronto

2026.01 – Present

Ongoing

- Building on “An Evidential Route to Asymptotic Bayes Optimality under Sparsity” (ICML 2026 poster), extended the relative-belief ABOS framework from additive 0-1 loss to general additive loss by deriving the loss-adjusted Bayes oracle, fixed-threshold ABOS window, and risk scale.
- Developed a regular scale-prior relative-belief threshold theory using local prior-predictive expansion, yielding ABOS conditions for one-group working priors.
- Unified normal and uniform relative-belief rules under the regular scale-prior framework and organized additional regular-prior cases including Laplace, Cauchy, and Student-t priors.

Survey-Based Health Risk Modeling with Interpretable Machine Learning

Advisor: Kun Tang, Associate Professor, Tsinghua University

2025.11 – Present

Intern Project

- Participated in questionnaire data cleaning and feature preparation for a large-scale student health survey, producing analysis-ready inputs for downstream modeling.
- Built and ran the full machine-learning pipeline, including imputation, stratified train/test evaluation, six-model benchmarking, class-imbalance handling, and interpretable model analysis with feature importance and SHAP outputs.

HONORS AND AWARDS

Freshman Scholarship, First Prize

Fudan University

2022

Basic Discipline Scholarship

Fudan University

2023–2024